Putting data on the Cloud (public or private) is an integral component of the IIoT. But this comes with huge security implications. Traditionally, industrial control system (ICS) vendors have maintained that their systems have a built-in air gap. This is no longer true when these systems have a direct or indirect connection to the Internet. The IIoT is going to drive the understanding that ICSes need to have embedded authentication and security features.

Let us now look at the network architecture that enables the IIoT. Fig. 3 provides a top-level view of how field devices in a factory or a manufacturing process are ultimately connected to the network.

There has always been a control network, a host of field sensors, actuators or servo drives (and other such devices) connected to PLCs or distributed control systems. Typically, this control network is a bunch of isolated networks. But increasingly, control networks that manage different sections of a factory or process are connected together, creating the plant network.

A plant network lets supervisors see the entire plant operation and deduce how the different sections of a plant interact with each other. Information at this level allows for optimisation of the entire plant or an oil field. Ultimately, this plant network information is integrated with the enterprise/business network to enable the real promise of the IIoT.

Each level of operation within the control network needs to have its security needs assessed—security is different at each level. If you start at the top, the domain of IT, you have secure switches and servers that are (hopefully) updated with the latest software and patches as explained below.

• At the plant level, security is not up to date. However, IT still has some control.

• At the control network layer, PLC architectures are decades old. Generally, updates are rare, and frequent patches cannot be applied to systems that are responsible for 100 per cent factory uptime. Security is generally weak here.

• At the field level, which is generally never discussed, security is virtually nonexistent. Field devices are open, trusted and cannot really have any encryption implemented because interoperability is paramount. If you look at field slave devices, such as sensors and actuators, these systems have zero security